

Learning Confidence Ellipsoids

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Motivation: Conformal Prediction

Strategy to ensure **reliability** of machine learning in practice

Prediction problem:

See $y_1, y_2, \dots, y_T$.

What is y_{T+1} ?

Classical answer:

Can learn a reasonable estimate $\hat{y}$, that has the **lowest expected error** with respect to some loss function

Issue:

- If best estimate $\hat{y}$ is going to be high error, want to know now!
- Evaluating error at the end of the game is too late to do anything about it
- Need **uncertainty quantification** on the fly $\longrightarrow$ consider **prediction sets**

Prediction/Confidence Sets

Prediction problem:

See $y_1, y_2, \dots, y_T$.

What is y_{T+1} ?

Conformal prediction:

For a given **coverage rate** $0 \leq (1 - \alpha) \leq 1$, predict a set $\mathcal{C}$, such that

$$\mathbb{P}_{y_1, \dots, y_T, y_{T+1}}[y_{T+1} \in \mathcal{C}] \geq 1 - \alpha,$$

and $\mathcal{C}$ is as small as possible.

$\implies$ study as a **constrained optimization** problem!

**Introduce Volume-
optimality Framework**

[Gao Shan S.
Vijayaraghavan ICML '25]

**Confidence Ellipsoids in
High-Dimensions**

[Gao Shan S.
Vijayaraghavan COLT '25,
arXiv '26]

This talk

**Online Conformal
Prediction with
Efficiency Guarantees**

[S. SODA '26]

Learning Confidence Ellipsoids

Problem: Given samples drawn i.i.d. from an unknown **arbitrary** distribution $\mathcal{D}$ over $\mathbb{R}^d$, and **coverage rate** $(1 - \alpha)$, fit **smallest volume** ellipsoid E such that

$$\mathbb{P}_{y \sim \mathcal{D}}[y \in E] \geq 1 - \alpha.$$

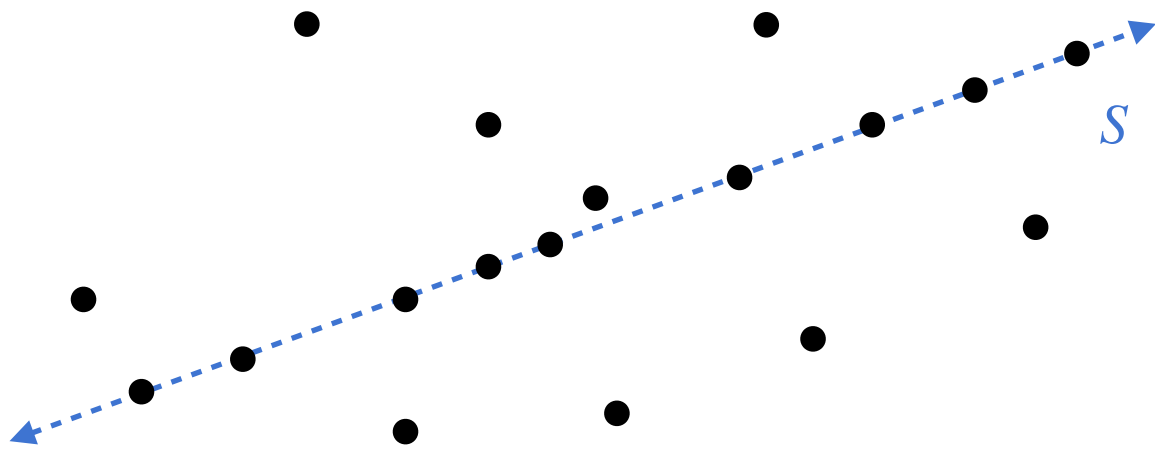
Focus on empirical problem over samples from $\mathcal{D}$ via **uniform convergence**

Applications: Central problem in statistics

- Rousseeuw's classical robust covariance estimator [Rousseeuw '84, 85],
- Conformal prediction [Gao Shan S. Vijayaraghavan '25],
- Estimating density level sets [Garcia Katalik Cho Wolkenhauer '03],
- Support estimation [Schölkopf Platt Shawe-Taylor Smola Williamson '01],
- and many more!

Challenge: Thin Ellipsoids

- Previous work [GSSV '25] gives a non-trivial volume approximation, when competing in volume against the best **ball**
- For thin ellipsoids, need to be able to handle NP-hard problem of **robust subspace recovery**



- Find hidden subspace S such that $(1 - \alpha)$ -fraction of points are on S
- Subspace contained in ellipsoid of volume 0
- Rules out any multiplicative volume approximation

- Aim for guarantee that depends on **condition number** β of ellipsoid E

$$\beta = \frac{\text{max. axis length of } E}{\text{min. axis length of } E} \geq 1$$

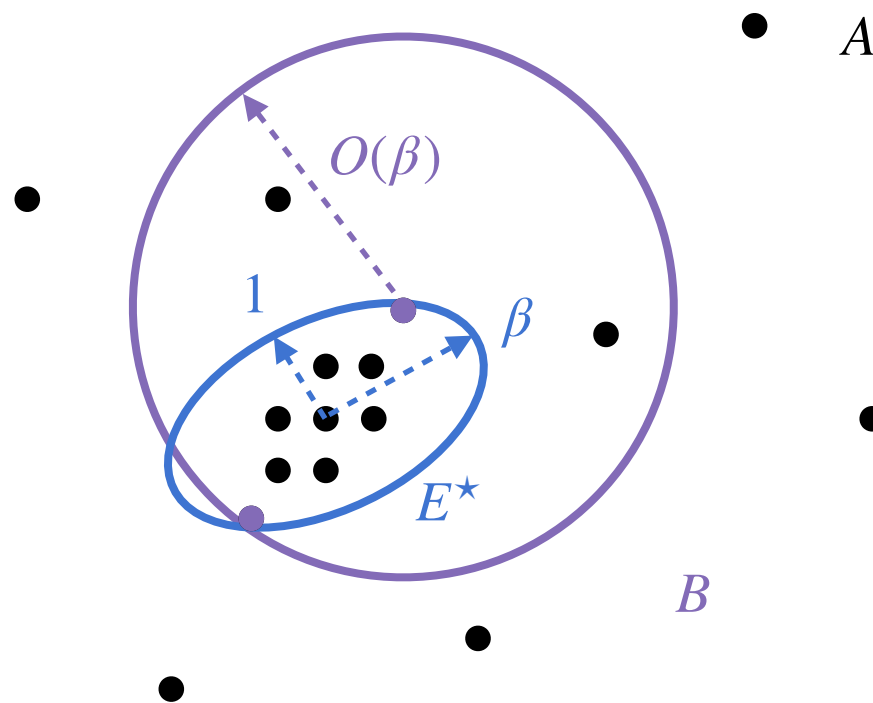
Sneak peak: guarantee with mild dependence on β

Condition Number Helps

- Let $E^\star$ be reference ellipsoid with condition number $\leq \beta$ and coverage $\geq 1 - \alpha$
- Can capture points in $E^\star$ in ball B with radius comparable to max. axis of $E^\star$

$$\text{radius}(B) \leq 2 \cdot \text{max. axis}(E^\star)$$

$$\implies \text{vol}(B) \leq \text{vol}(E^\star) \cdot O(\beta)^d$$



Goal: nontrivial approximation better than β^d

Result

Theorem. Let $\alpha > 0$ be a miscoverage rate, $\gamma > 0$ a parameter, and $\beta \geq 1$ a condition number bound.

Let $A = \{a_1, \dots, a_n\} \subseteq \mathbb{R}^d$ be a set of points, and $E^\star$ be an arbitrary ellipsoid of condition number at most β such that $|E^\star \cap A| \geq (1 - \alpha)n$.

We give a polynomial-time algorithm that finds an ellipsoid $\hat{E}$ such that

$$\text{vol}(\hat{E}) \leq \text{vol}(E^\star) \cdot \beta^{O(\gamma d)},$$

And the coverage of $\hat{E}$ is bounded as

$$|\hat{E} \cap A| \geq \left(1 - \frac{c\alpha}{\gamma}\right)n,$$

for a constant c .

- Think of $\gamma = \sqrt{\alpha}$, get coverage $(1 - \sqrt{\alpha})$, volume approx. $\exp(\sqrt{\alpha} \cdot d \log \beta)$
- **Tight** via SSE-hardness of robust subspace recovery
- Gives first algorithm for **robust subspace recovery** on arbitrary points

Minimum Volume Enclosing Ellipsoid

For talk: focus on origin-centric enclosing ellipsoid (full details in paper)

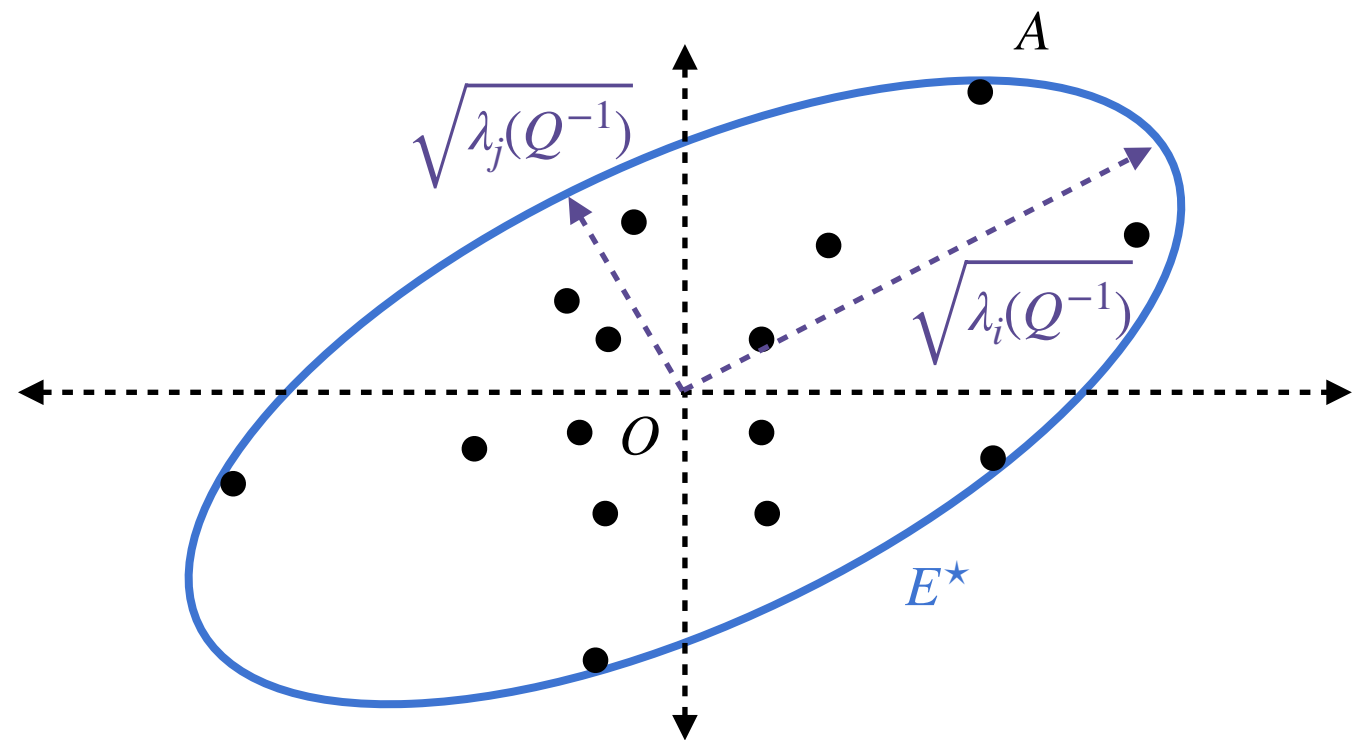
How to enclose all points $A \subseteq \mathbb{R}^d$? ($\alpha = 0$)

SDP for John Ellipsoid

$$\begin{aligned} & \max_Q \log \det Q \\ & \text{s.t. } \forall a \in A, a^\top Q a \leq 1 \\ & \quad Q \geq 0 \end{aligned}$$

$$E^\star = \{x \in \mathbb{R}^d : x^\top Q x \leq 1\}$$

$$\text{vol}(E^\star) = \kappa_d \cdot (\det(Q^{-1}))^{1/2}$$



Miscoverage: only want to cover $(1 - \alpha)$ -fraction of points A

Goal: remove points from A that increase volume of $E^\star$ the most

Insight: SDP must identify these points, since they are hardest to cover!

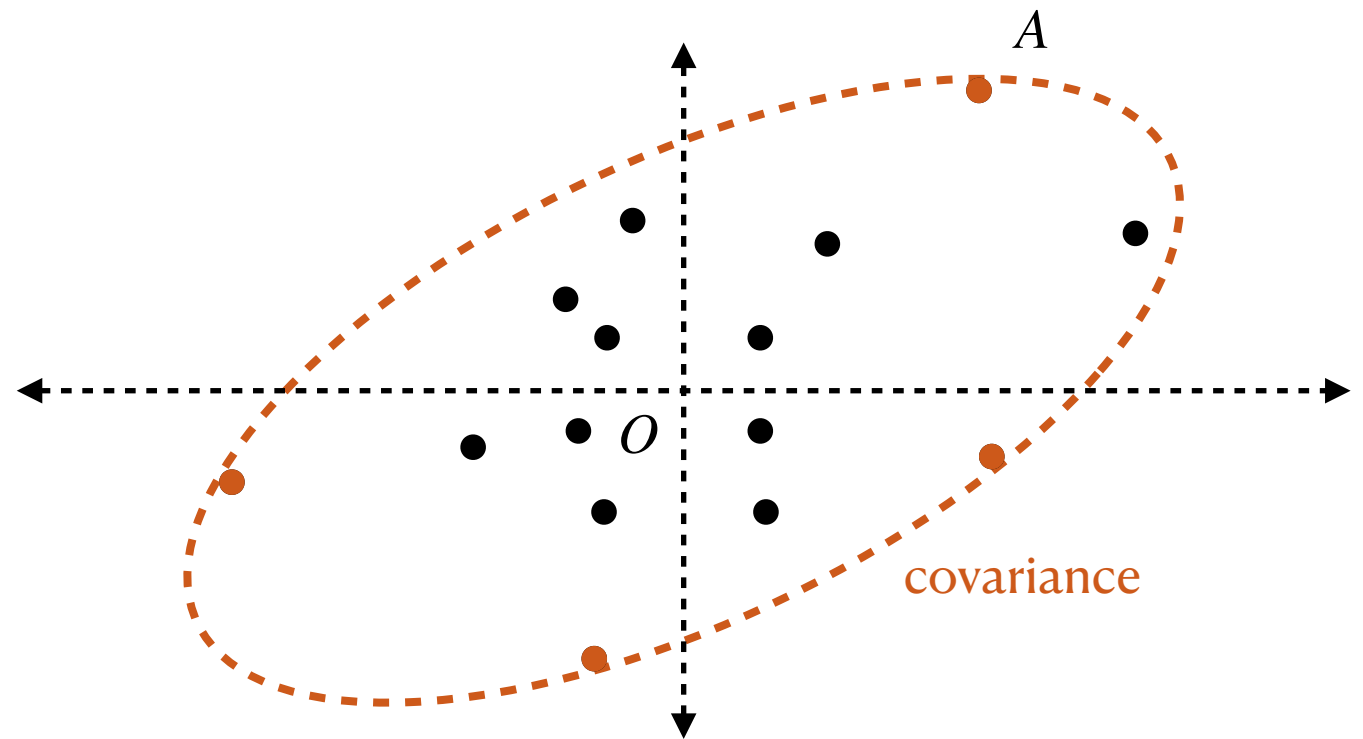
Primal SDP

$$\begin{aligned} \max_Q \quad & \log \det Q \\ \text{s.t.} \quad & \forall a \in A, a^\top Q a \leq 1 \\ & Q \geq 0 \end{aligned}$$

Dual Formulation

Dual SDP

$$\begin{aligned} \max_{w_i, \forall a_i \in A} \quad & \log \det \left(\sum_{i=1}^n w_i a_i a_i^\top \right) \\ \text{s.t.} \quad & \forall i, w_i \geq 0 \\ & \sum_{i=1}^n w_i \leq d \end{aligned}$$



(each $w_i \leq 1$, WLOG)

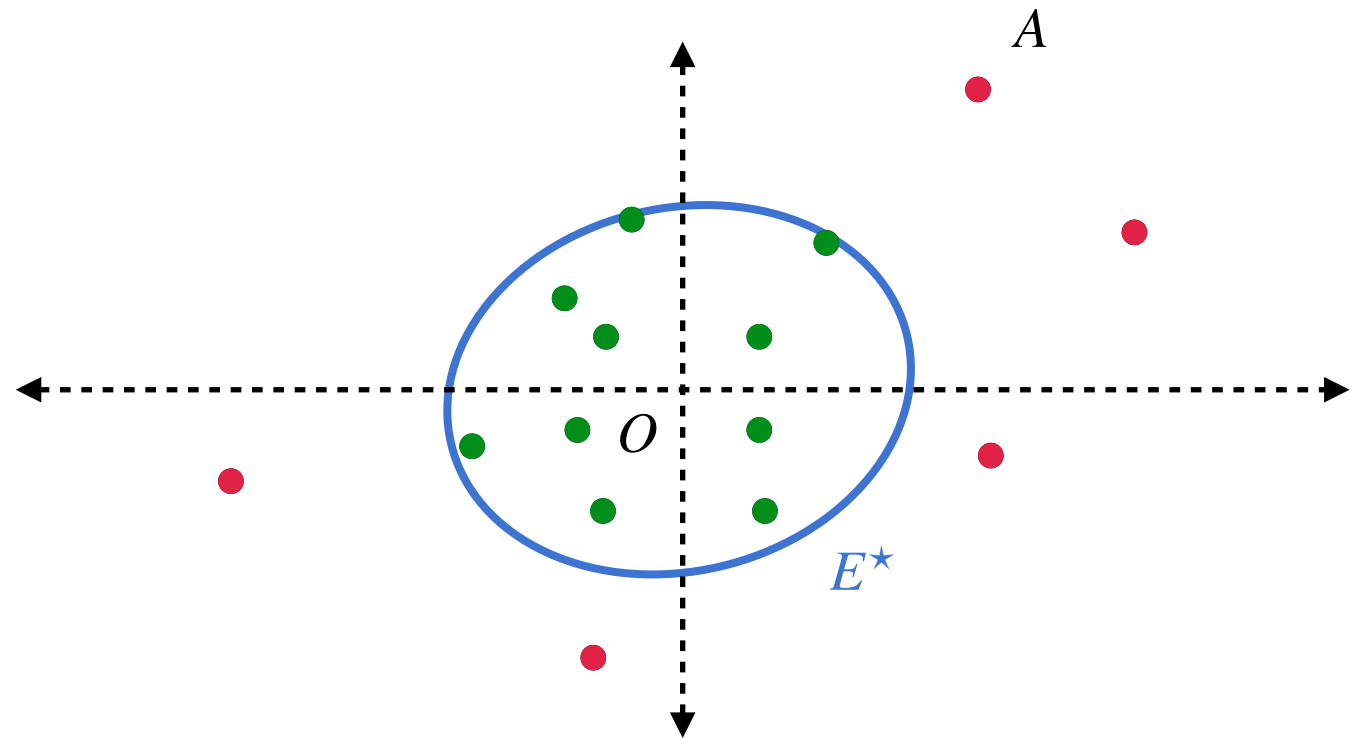
$$\sum_{i=1}^n w_i a_i a_i^\top = M = Q^{-1}$$

Intuition:

- If a high-variance distribution is supported on A , then A must be spread out
- **Complementary slackness:** only points on boundary of ellipsoid have $w_i > 0$
 $\iff$ min. enclosing ellipsoid of A = min. enclosing ellipsoid of a_i s with $w_i > 0$
- Think of w_i a measure of how important a_i is to volume of enclosing ellipsoid
- **Idea:** remove points with large w_i values

Removing outliers

- $E^\star$: true min.-vol. ellipsoid of condition number $\leq \beta$ containing $(1 - \alpha)$ -fraction of A
- **Inliers**: $A \cap E^\star$
- **Outliers**: $A \setminus E^\star$



Strategy:

- Solve **dual program** on included points a_i to get weights w_i
- Remove each point a_i from A with probability w_i
- Iterate

Let $0 < \gamma < 1$ be a parameter.

- If $\geq \gamma$ fraction of weight w_i on **outliers** $\implies$ progress removing **outliers** with $O(1/\gamma)$ blow up in # points removed
- **Claim**: if $< \gamma$ fraction of weight w_i on **outliers**, enclosing ellipsoid of A already a good approximation of $E^\star$!

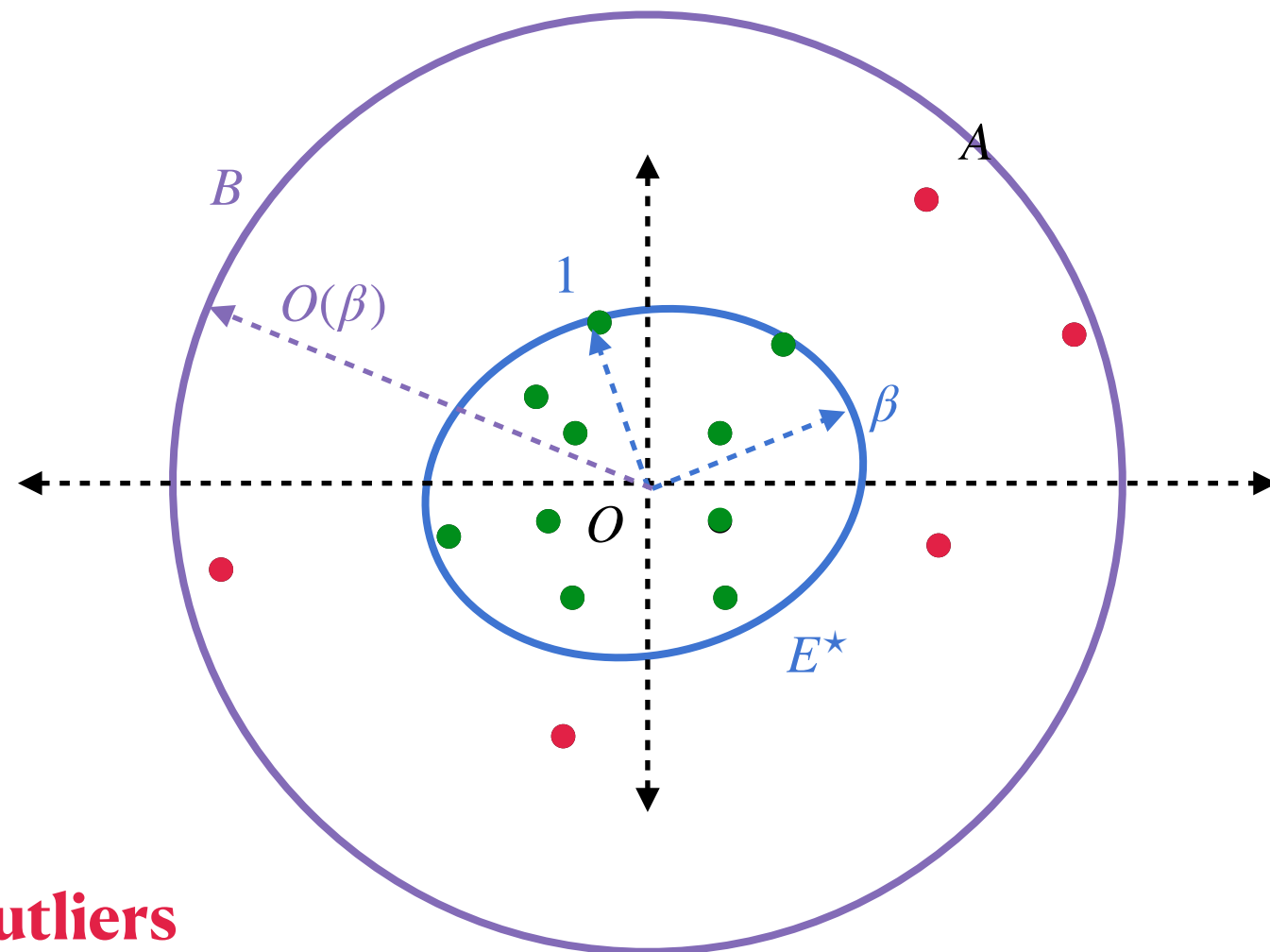
Proof of Concept

Claim: if $< \gamma$ fraction of weight w_i on **outliers**, enclosing ellipsoid of A already a good approximation of $E^\star$!

Recall: weights $w_i \in [0,1]$, $\sum_i w_i \leq d$

Assume for a moment: weights $w_i \in \{0,1\}$

- Rescale space so that $E^\star$ has width ≥ 1 in every direction
- Because of condition number bound, $E^\star$ has width $\leq \beta$ in every direction
- Capture points in ball B of radius $O(\beta)$
- **Goal:** enclose support of w_i , with $\leq \gamma d$ **outliers**
- Capture all **inliers** with $E^\star$, stretch by $\leq \beta$ for each **outlier**
- $\implies$ Exists an ellipsoid enclosing support of w_i , with volume $\leq \text{vol}(E^\star) \cdot \beta^{\gamma d}$

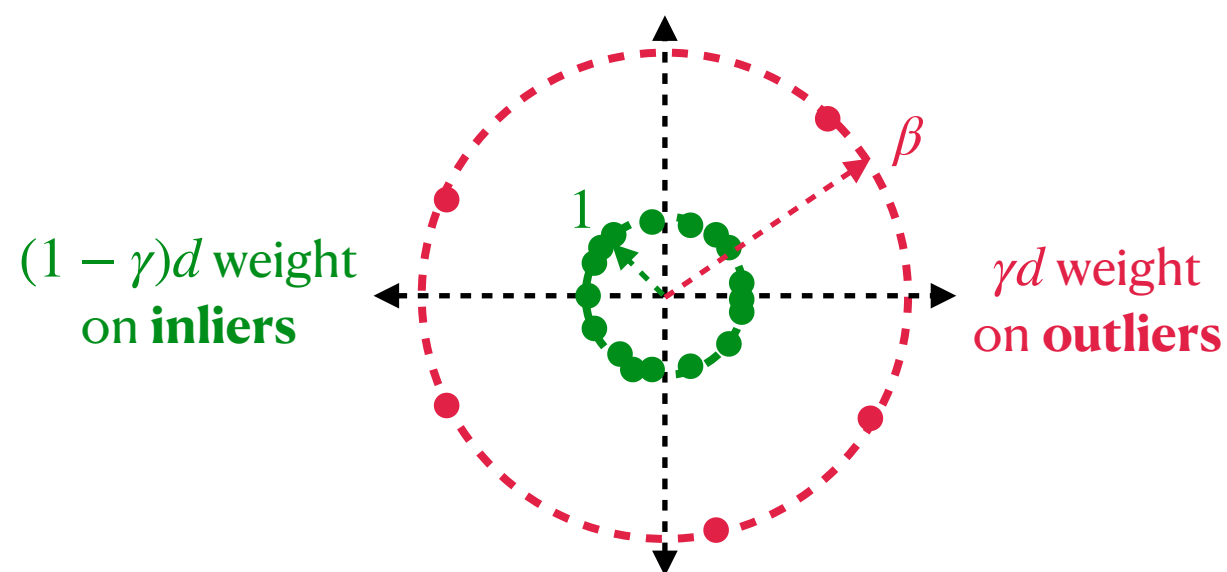


Challenge: spread outliers

Claim: if $< \gamma$ fraction of weight w_i on **outliers**, enclosing ellipsoid of A already a good approximation of $E^\star$!

What about non-integer weights $w_i \in [0,1]$?

Previous argument used that not only was there $\leq \gamma d$ weight on **outliers**, but that it was spread in $\leq \gamma d$ directions



$$\text{Cov} = (1 - \gamma) \cdot I_d + \gamma \cdot \beta I_d \\ \approx (1 + \gamma\beta) \cdot I_d$$

$$\det(\text{Cov}) \approx (1 + \gamma\beta)^d = \exp(\gamma d \cdot \beta)$$

$$\det(\text{Cov}) \text{ of inliers} \approx 1$$

better
dependence
on β ?

- But optimal SDP solution would never pick this! $\implies$ Optimal solution would place more weight on **outliers** if they brought up the variance
- **Want to show:** if optimal solution places $\leq \gamma d$ weight on **outliers**, they are effectively only in $\leq \gamma d$ directions

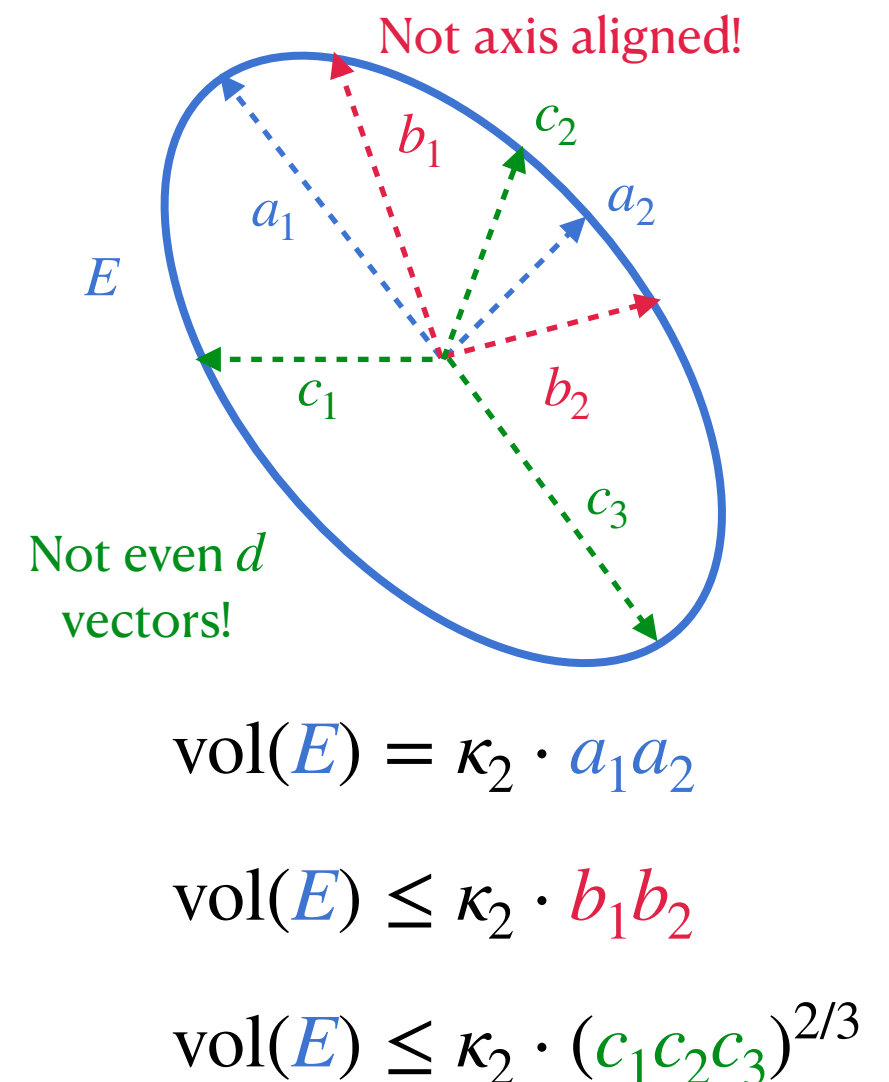
Geometric Brascamp-Lieb

Theorem [Ball '89] For any PSD matrix $P \in \mathbb{R}^{d \times d}$, and weights $w_i > 0$ and unit vectors $\|b_i\| = 1$, such that $\sum_i w_i b_i b_i^\top = I_d$,

$$\log \det(P) \leq \sum_i w_i \log(b_i^\top P b_i).$$

Intuition:

- b_i s form a **tight frame**, generalization of a basis
- When b_i s are eigenvectors of P , with $w_i = 1$, then
$$\sum_i w_i \log(b_i^\top P b_i) = \sum_i \log(\lambda_i(P)) = \log \det(P)$$
- Eigenbasis is the most efficient way to measure determinant, any other basis will only give you a bigger estimate
- Can prove through concavity of log



Bounding Volume Approximation

- Let $M = \sum_i w_i a_i a_i^\top$
- $E^\star = \{x : x^\top S^{-1} x \leq 1\}$ optimal ellipsoid enclosing **inliers**.
- Let $P = M^{1/2} S^{-1} M^{1/2}$. So,

$$\det(M) = \det(S) \cdot \det(P).$$

$$\text{volume approximation} \approx \det(P)$$

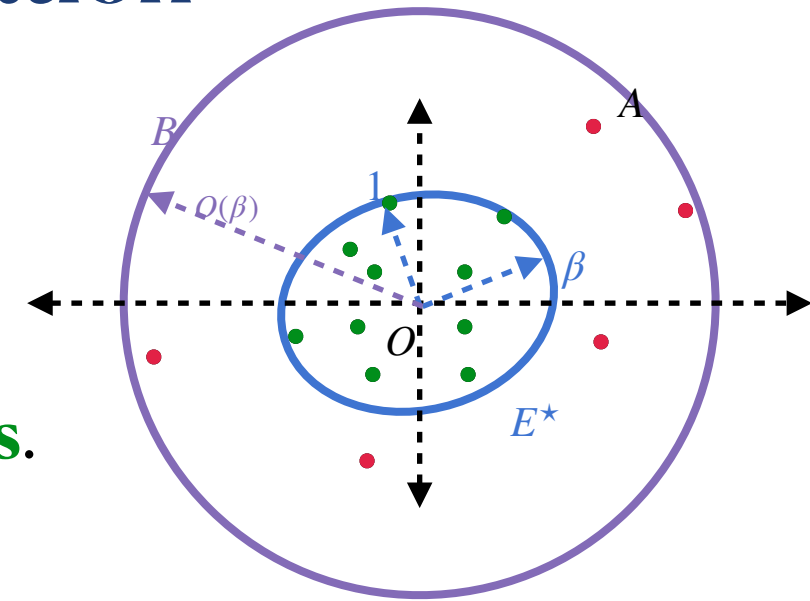
- Hard to figure out the eigenvectors/values of P to measure volume. Instead, measure according to points a_i using **geometric Brascamp-Lieb**!

$$\log \det(P) \leq \sum_i w_i \log(a_i^\top S^{-1} a_i)$$

- Bound length along a_i as ≤ 1 for **inliers** and $\leq \beta$ for **outliers**

$$\log \det(P) \leq O(\gamma d) \cdot \log \beta$$

$$\det(P)^{1/2} \leq \beta^{O(\gamma d)}$$



Compare to $\exp(\gamma d \cdot \beta)$
from naive bound

Result Again

Theorem. Let $\alpha > 0$ be a miscoverage rate, $\gamma > 0$ a parameter, and $\beta \geq 1$ a condition number bound.

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And the coverage of $\hat{E}$ is bounded as

$$|\hat{E} \cap A| \geq \left(1 - \frac{c\alpha}{\gamma}\right)n,$$

for a constant c .

Conclusion

In this talk:

- New algorithm to find **confidence ellipsoids** of approximately optimal volume
- New insights into **primal-dual structure** of the SDP for minimum volume enclosing ellipsoid

Not in the talk (see paper!):

- Our approximation factor is tight (up to constants in the exponent) via SSE-hardness of **robust subspace recovery**
- First algorithm for **robust subspace recovery** on arbitrary points

(Paper title: Learning Confidence Ellipsoids and Applications to Robust Subspace Recovery)

Thanks!

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